

In []:

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"""
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_____STUDENT GRADE ANALYZE PROJECT_____
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Description:
This project analyzes student performance using a real dataset.
It includes:
- Data Loading from CSV
- Data Inspection & Cleaning
- Feature Engineering (Average Score)
- Grade Classification
- Insights Generation
- Data Visualization
- Exporting Results

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"""

# =====
#   IMPORT LIBRARIES
# =====
import pandas as pd
import matplotlib.pyplot as plt

# =====
#   LOAD DATA
# =====
def load_data(Student):
    df = pd.read_csv("C:/Users/Mubeen/Desktop/Student-Record.csv")
    return df

# =====
#   DATA INSPECTION
# =====
def inspect_data(df):
    print("\n FIRST 5 ROWS:")
    print(df.head())

    print("\n DATA INFO:")
    print(df.info())

    print("\n MISSING VALUES:")
    print(df.isnull().sum())

# =====
#   DATA CLEANING
# =====
def clean_data(df):
    # Remove duplicates
    df = df.drop_duplicates()

    # Handle missing values (if any)
    df = df.dropna()
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    return df

# =====
#     FEATURE ENGINEERING
# =====
def add_average(df):
    df["Average"] = df[["Math", "English", "Science"]].mean(axis=1).round(2)
    return df

def assign_grade(avg):
    if avg >= 90:
        return "A"
    elif avg >= 80:
        return "B"
    elif avg >= 70:
        return "C"
    elif avg >= 60:
        return "D"
    else:
        return "F"

def add_grades(df):
    df["Grade"] = df["Average"].apply(assign_grade)
    return df

# =====
#     ANALYSIS
# =====
def get_top_student(df):
    return df.loc[df["Average"].idxmax()]

def generate_insights(df):
    print("\n KEY INSIGHTS:")

    print(f"Average Class Score: {df['Average'].mean():.2f}")

    subject_avg = df[["Math", "English", "Science"]].mean()
    print(f"Best Performing Subject: {subject_avg.idxmax()}")
    print(f"Weakest Subject: {subject_avg.idxmin()}")

    print(f"Highest Score: {df['Average'].max()}")
    print(f"Lowest Score: {df['Average'].min()}")

# =====
#     VISUALIZATION (MANDATORY)
# =====
def plot_grade_distribution(df):
    grade_counts = df["Grade"].value_counts().sort_index()

    plt.figure()
    grade_counts.plot(kind="bar")
    plt.title("Grade Distribution")
    plt.xlabel("Grades")

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plt.ylabel("Number of Students")
plt.show()

def plot_subject_averages(df):
    subject_avg = df[["Math", "English", "Science"]].mean()

    plt.figure()
    subject_avg.plot(kind="bar")
    plt.title("Average Score per Subject")
    plt.xlabel("Subjects")
    plt.ylabel("Average Marks")
    plt.show()

def plot_score_distribution(df):
    plt.figure()
    df["Average"].plot(kind="hist", bins=10)
    plt.title("Distribution of Average Scores")
    plt.xlabel("Average Score")
    plt.ylabel("Frequency")
    plt.show()

# =====
#         EXPORT RESULTS
# =====

def export_results(df):
    df.to_csv("final_student_report.csv", index=False)
    print("\n Report exported as 'final_student_report.csv'")

# =====
#         MAIN FUNCTION
# =====

def main():
    file_path = "students_dataset_120.csv" # Your dataset file

    # Step 1: Load Data
    df = load_data(file_path)

    # Step 2: Inspect Data
    inspect_data(df)

    # Step 3: Clean Data
    df = clean_data(df)

    # Step 4: Feature Engineering
    df = add_average(df)
    df = add_grades(df)

    # Step 5: Analysis
    top_student = get_top_student(df)

    print("\n TOP PERFORMER:")
    print(top_student)

    # Step 6: Insights
    generate_insights(df)

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# Step 7: Visualization
plot_grade_distribution(df)
plot_subject_averages(df)
plot_score_distribution(df)

# Step 8: Export
export_results(df)

# =====
#          RUN PROJECT
# =====
if __name__ == "__main__":
    main()

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FIRST 5 ROWS:

	Name	Math	English	Science
0	Student_1	53	51	67
1	Student_2	66	96	69
2	Student_3	73	73	89
3	Student_4	58	70	76
4	Student_5	96	61	53

DATA INFO:

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<class 'pandas.core.frame.DataFrame'>
RangeIndex: 120 entries, 0 to 119
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Name        120 non-null   object
1   Math        120 non-null   int64
2   English     120 non-null   int64
3   Science     120 non-null   int64
dtypes: int64(3), object(1)
memory usage: 3.9+ KB
None

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MISSING VALUES:

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Name      0
Math      0
English   0
Science   0
dtype: int64

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TOP PERFORMER:

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Name      Student_8
Math      94
English   92
Science   95
Average   93.67
Grade     A
Name: 7, dtype: object

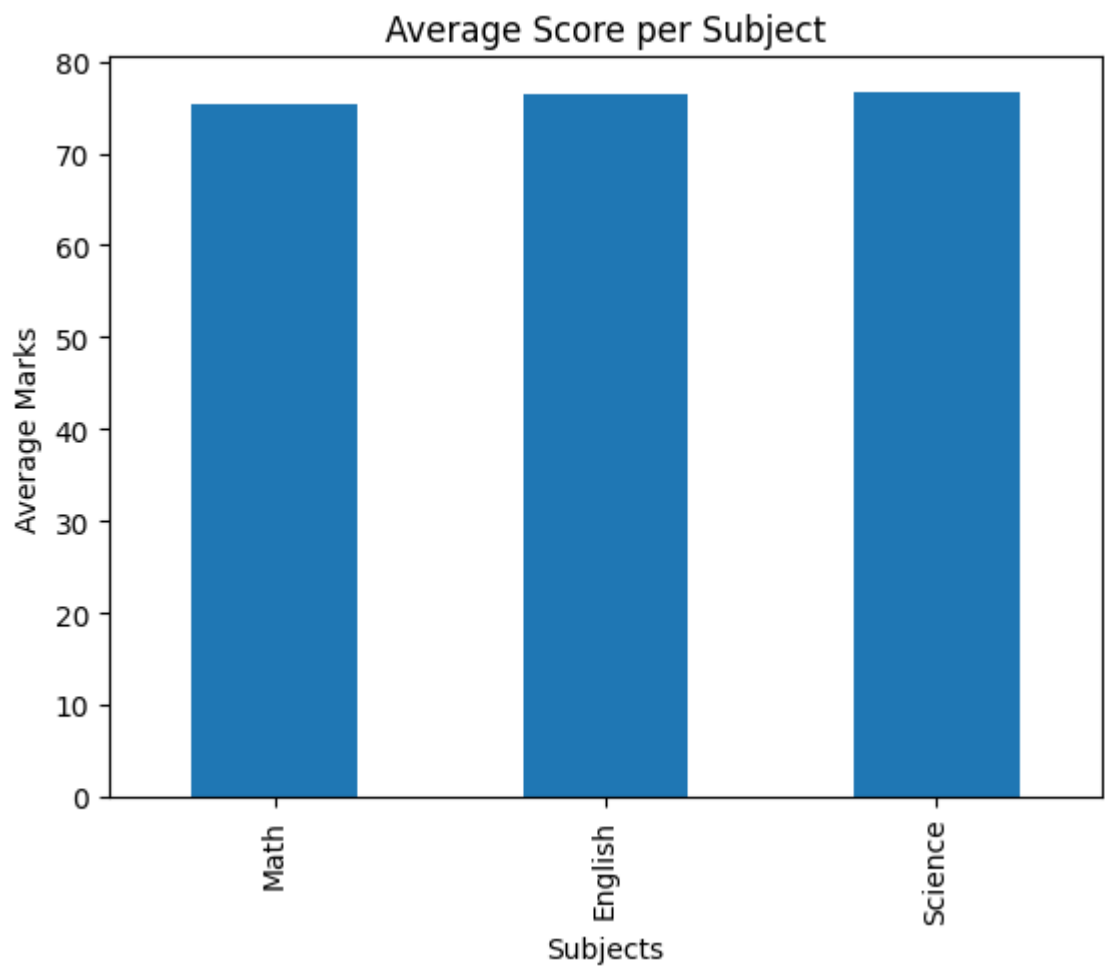
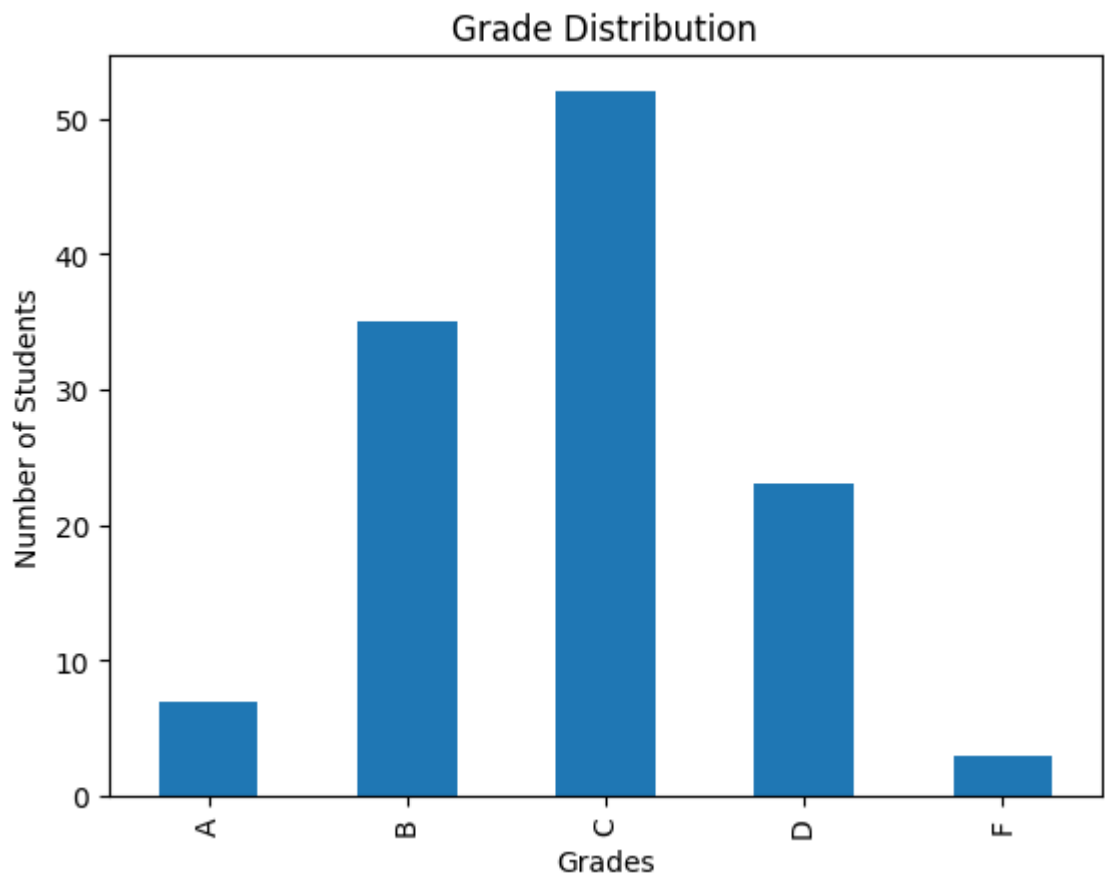
```

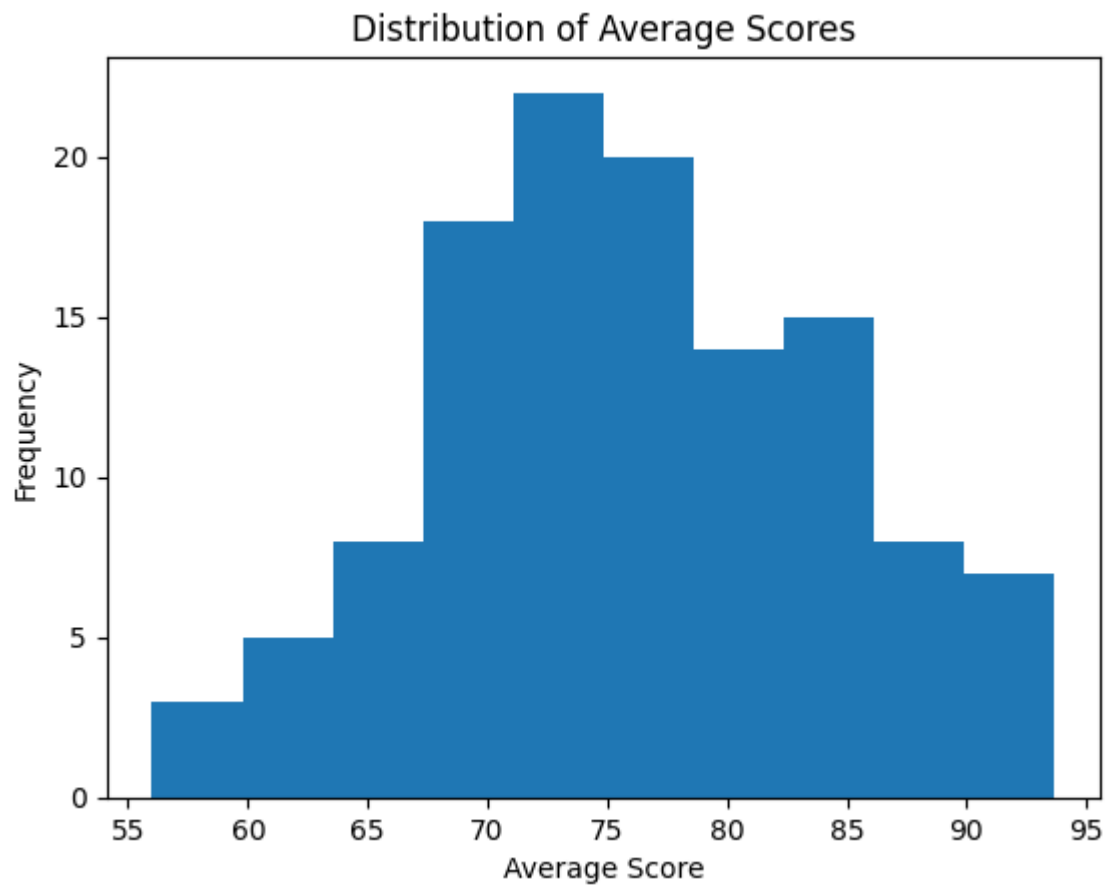
KEY INSIGHTS:

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Average Class Score: 76.14
Best Performing Subject: Science
Weakest Subject: Math
Highest Score: 93.67
Lowest Score: 56.0

```





Report exported as 'final_student_report.csv'